IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

Maximum Likelihood Estimation - I

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT STUDIES

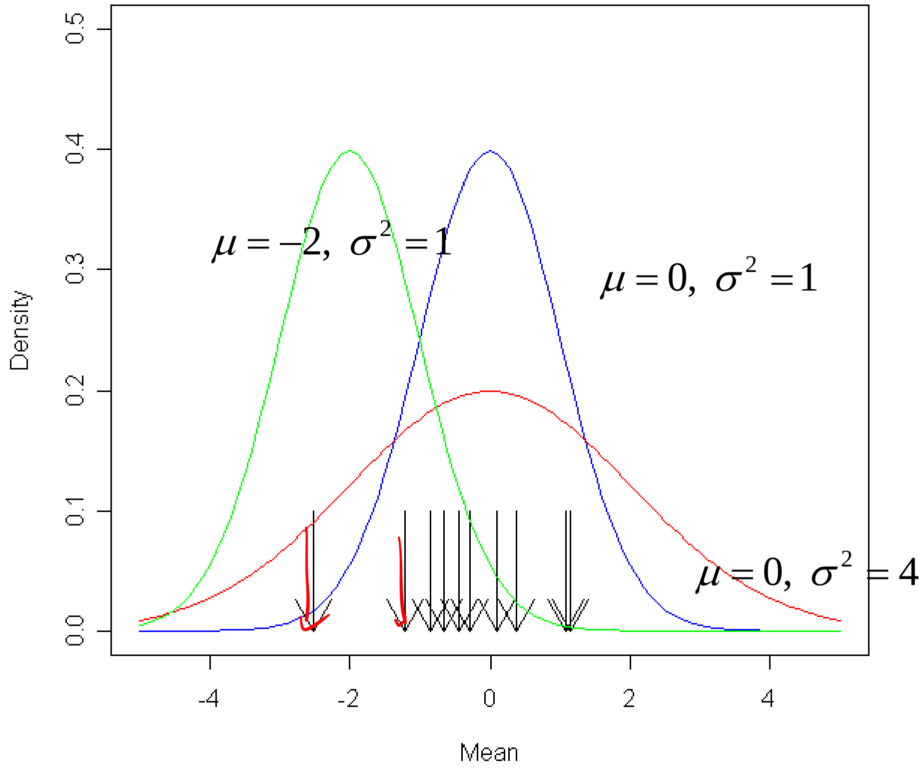
Agenda

- This lecture will provide intuition behind the MLE using Theory and examples.

Maximum Likelihood Estimation

- The method of maximum likelihood was first introduced by R. A. Fisher, a geneticist and statistician, in the 1920s.
- Most statisticians recommend this method, at least when the sample size is large, since the resulting estimators have certain desirable efficiency properties
- Maximum likelihood estimation(MLE) is a method to find most likely density function, that would have generated data.
- MLE requires one to make distribution assumption first.

An intuitive view on likelihood



Maximum Likelihood Estimation: Problem

- A sample of ten new bike helmets manufactured by a certain company is obtained. Upon testing, it is found that the first, third, and tenth helmets are flawed, whereas the others are not.
- Let p = P(flawed helmet), i.e., p is the proportion of all such helmets that are flawed.
- Define (Bernoulli) random variables $X_1, X_2, \ldots, X_{10}$ by

$$X_1 = \begin{cases} 1 \text{ if 1st helmet is flawed} \\ 0 \text{ if 1st helmet isn't flawed} \end{cases} \quad \cdots \quad X_{10} = \begin{cases} 1 \text{ if 10th helmet is flawed} \\ 0 \text{ if 10th helmet isn't flawed} \end{cases}$$

Source: Probability and Statistics for Engineering and the Sciences, Jay L Devore, 8th Ed, Cengage

Maximum Likelihood Estimation: Problem

- Then for the obtained sample, $X_1 = X_3 = X_{10} = 1$ and the other seven X_i's are all zero
- The probability mass function of any particular X_i is $p^{x_i}(1-p)^{1-x_i}$, which becomes p if $x_i = 1$ and $1 - p$ when $x_i = 0$
- Now suppose that the conditions of various helmets are independent of one another
- This implies that the X_i's are independent, so their joint probability mass function is the product of the individual pmf's.

Maximum Likelihood Estimation: Binomial Distribution

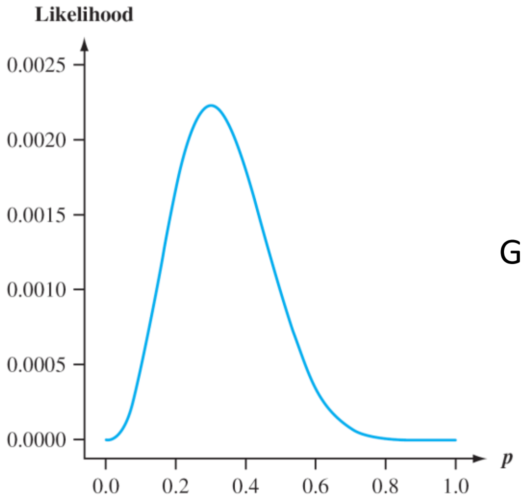
- Joint pmf evaluated at the observed X_i's is

  $f(x_1, \ldots, x_{10}; p) = p(1-p)p \cdots p = p^3(1-p)^7$ - (1)

- Suppose that $p = .25$. Then the probability of observing the sample that we actually obtained is $(.25)^3(.75)^7 = .002086$.
- If instead $p = .50$, then this probability is $(.50)^3(.50)^7 = .000977$.
- For what value of p is the obtained sample most likely to have occurred?
- That is, for what value of p is the joint pmf (eq 1) as large as it can be?
- What value of p maximizes (eq 1)

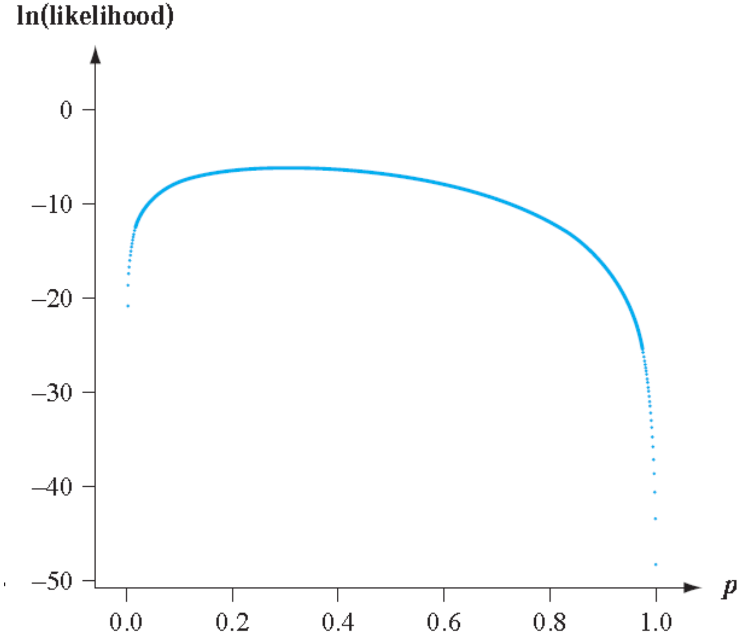
Maximum Likelihood Estimation: Binomial Distribution

- Figure shows a graph of the *likelihood* (eq 1) as a function of p.
- It appears that the graph reaches its peak above p = .3 = the proportion of flawed helmets in the sample.



Graph of the likelihood (joint pmf) (eq 1)

Graph of the natural logarithm of the likelihood



- Figure shows a graph of the natural logarithm of (eq 1)
- Since *ln*[g(u)] is a strictly increasing function of $g(u)$, finding u to maximize the function $g(u)$ is the same as finding u to maximize *ln*[g(u)].

Maximum Likelihood Estimation: Binomial Distribution

- We can verify our visual impression by using calculus to find the value of p that maximizes (eq 1).
- Working with the natural log of the joint pmf is often easier than working with the joint pmf itself, since the joint pmf is typically a product so its logarithm will be a sum.
- Here $\ln[f(x_1, \ldots, x_{10}; p)] = \ln[p^3(1-p)^7]$
- $= 3\ln(p) + 7\ln(1-p)$

Maximum Likelihood Estimation: Binomial Distribution

Thus $$\frac{d}{dp}\{\ln[f(x_1, \ldots, x_{10}; p)]\} = \frac{d}{dp}\{3\ln(p) + 7\ln(1-p)\}$$

$$= \frac{3}{p} + \frac{7}{1-p}(-1)$$

$$= \frac{3}{p} - \frac{7}{1-p}$$

Interpretation

- Equating this derivative to 0 and solving for p gives
  $3(1 - p) = 7p$, from which $3 = 10p$ and so $p = 3/10 = .30$ as conjectured

- That is, our point estimate is $p = .30$.
- It is called the *maximum likelihood estimate* because it is the parameter value that maximizes the likelihood (joint pmf) of the observed sample
- In general, the second derivative should be examined to make sure a maximum has been obtained, but here this is obvious from Figure

Maximum Likelihood Estimation: Binomial Distribution

- Suppose that rather than being told the condition of every helmet, we had only been informed that three of the ten were flawed.
- Then we would have the observed value of a binomial random variable X = the number of flawed helmets.
- The pmf of X is $\binom{10}{x} p^x (1 - p)^{10-x}$. For x = 3, this becomes $\binom{10}{3} p^3 (1 - p)^7$.
- The binomial coefficient $\binom{10}{3}$ is irrelevant to the maximization, so again p = 0.30.

Maximum Likelihood Function Definition

- Let $X_1, X_2,\ldots, X_n$ have joint pmf or pdf

$$f(x_1, x_2, \ldots, x_n; \theta_1, \ldots, \theta_m) \qquad \text{(a)}$$

- Where the parameters $\theta_1, \ldots, \theta_m$ have unknown values. When $x_1, \ldots, x_n$ are the observed sample values and (a) is regarded as a function of $\theta_1, \ldots, \theta_m$, it is called the **likelihood function.**

- The maximum likelihood estimates (mle's) $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are those values of the *i*'s that maximize the likelihood function, so that

$$f(x_1, x_2, \ldots, x_n; \hat{\theta}_1, \ldots, \hat{\theta}_m) \geq f(x_1, x_2, \ldots, x_n; \theta_1, \ldots, \theta_m) \text{ for all } \theta_1, \ldots, \theta_m$$

- When the $X_i' s$ are substituted in place of the $x_i' s$, the **maximum likelihood estimators** result.

Interpretation

- The likelihood function tells us how likely the observed sample is as a function of the possible parameter values.
- Maximizing the likelihood gives the parameter values for which the observed sample is most likely to have been generated—that is, the parameter values that “agree most closely” with the observed data.

Estimation of Poisson Parameter

- Suppose we have data generated from a Poisson distribution. We want to estimate the parameter of the distribution
- The probability of observing a particular random variable is $P(X;\mu) = \dfrac{e^{-\mu}\mu^{X}}{X!}$
- Joint likelihood by multiplying the individual probabilities together

$$P(X_1, X_2, \ldots, X_n; \mu) = \frac{e^{-\mu}\mu^{X_1}}{X_1!} \times \frac{e^{-\mu}\mu^{X_2}}{X_2!} \times \ldots \times \frac{e^{-\mu}\mu^{X_n}}{X_n!}$$

$$L(\mu; \mathbf{X}) = \prod_i e^{-\mu}\mu^{X_i}$$

$$L(\mu; \mathbf{X}) = e^{-n\mu}\mu^{n\overline{X}}$$

Estimation of Poisson Parameter

- Note in the likelihood function the factorials have disappeared.
- This is because they provide a constant that does not influence the relative likelihood of different values of the parameter
- It is usual to work with the **log likelihood** rather than the likelihood.
- Note that maximising the log likelihood is equivalent to maximising the likelihood.

$$L(\mu;\mathbf{X}) = e^{-n\mu}\mu^{n\bar{X}}$$

Take the natural log of the likelihood function

$$\ell(\mu;\mathbf{X}) = -n\mu + n\bar{X}\log\mu$$

Find where the derivative of the log likelihood is zero

$$\frac{d\ell}{d\mu} = -n + \frac{n\bar{X}}{\mu}$$

$$\hat{\mu} = \bar{X}$$

Note that here the MLE is the same as the moment estimator

Estimation of exponential distribution Parameter

- Suppose $X_1, X_2, \ldots, X_n$ is a random sample from an exponential distribution with parameter λ. Because of independence, the likelihood function is a product of the individual pdf's:

$$f(x_1, \ldots, x_n; \lambda) = (\lambda e^{-\lambda x_1}) \cdot \cdots \cdot (\lambda e^{-\lambda x_n})$$
$$= \lambda^n e^{-\lambda \Sigma x_i}$$

- The natural logarithm of the likelihood function is

$$\ln[f(x_1, \ldots, x_n; \lambda)] = n \ln(\lambda) - \lambda \Sigma x_i$$

Estimation of exponential distribution Parameter

- Equating $(d/d\lambda)[\ln(\text{likelihood})]$ to zero results in $n/\lambda - \Sigma x_i = 0$, or $\lambda = n/\Sigma x_i = 1/\bar{x}$.

- Thus the MLE is $\hat{\lambda} = 1/\bar{X}$;

Estimation of parameters of Normal Distribution

- Let $X_1, \ldots, X_n$ be a random sample from a normal distribution.
- The likelihood function is

$$f(x_1, \ldots, x_n; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_1-\mu)^2/(2\sigma^2)} \cdot \ldots \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_n-\mu)^2/(2\sigma^2)}$$

$$= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} e^{-\Sigma(x_i-\mu)^2/(2\sigma^2)}$$

- so

$$\ln[f(x_1, \ldots, x_n; \mu, \sigma^2)] = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

Estimation of parameters of normal distribution

- To find the maximizing values of μ and σ^2, we must take the partial derivatives of ln(f) with respect to μ and σ^2, equate them to zero, and solve the resulting two equations.

- Omitting the details, the resulting MLE's are

$$\hat{\mu} = \overline{X} \qquad \hat{\sigma}^2 = \frac{\sum(X_i - \overline{X})^2}{n}$$

- The MLE of σ^2 is not the unbiased estimator, so two different principles of estimation (unbiasedness and maximum likelihood) yield two different estimators

Thank you

IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Maximum Likelihood Estimation-II

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- This lecture will provide understanding intuition behind the MLE using Theory and examples.

Example1: **Estimation of parameters of normal distribution**

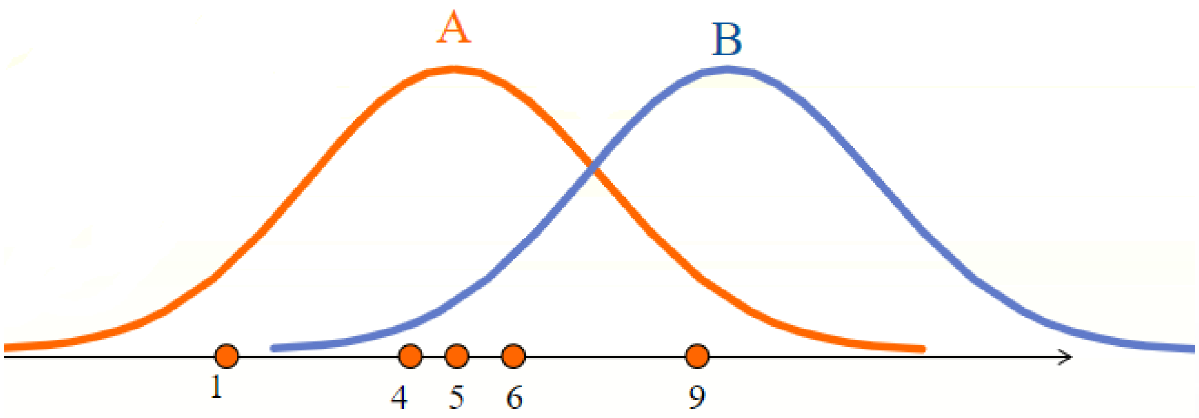
- Let us explain basic idea of MLE using simple problems.
- Let us make assumption that variable x follows normal distributed
- Density function of normal distribution with mean μ and variance σ^2 is given by:

Id	x
1	1
2	4
3	5
4	6
35	9

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} \quad \text{for} \quad -\infty < x < \infty$$

Example 1: **Estimation of parameters of normal distribution**

- The data is plotted on a horizontal line
- Think which distribution, either A or B, is more likely to have generated the data?



Interpretation

- Answer to this question is A, because the data are cluster around the center of the distribution A, but not around the center of the distribution B
- This example illustrate that, by looking at the data, it is possible to find the distribution that is most likely to have generated the data
- Now, I will explain exactly how to find the distribution in practice

The illustration of the estimation procedure

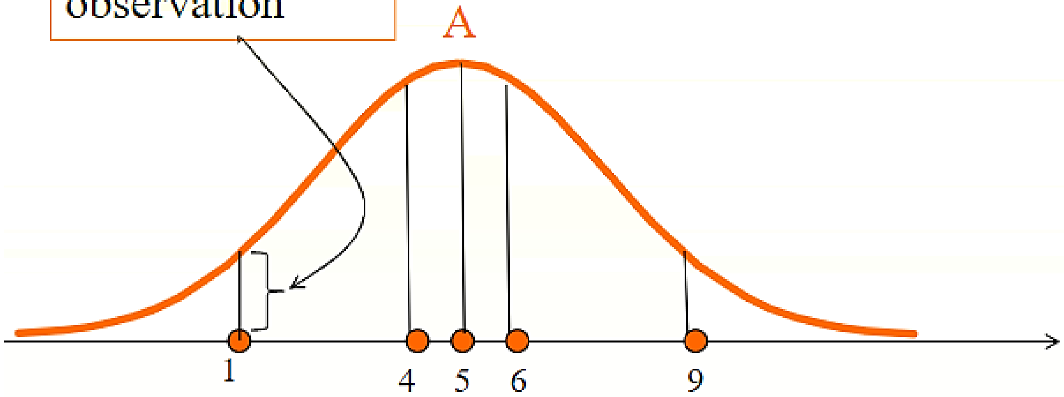
- MLE starts with computing the likelihood contribution of each observation
- The likelihood contribution is the height of the density function.
- We use L_i to denote the likelihood contribution of i^{th} observation.

Graphical illustration of likelihood contribution

The likelihood contribution of the first observation

$$= L_1 = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1-\mu)^2/2\sigma^2}$$

Data value



The illustration of the estimation procedure

- Then, you multiply the likelihood contributions of all the observations. this is called the likelihood function. We use the notation L
- Likelihood function L= $\prod_{i=1}^{n} L_i$

This notation means you multiply from i= 1 through n

- In our example, n= 5

The illustration of the estimation procedure

- In our example, the likelihood function looks like:

$$L(\mu, \sigma) = \prod_{i=1}^{5} L_i = L_1 \times L_2 \times L_3 \times L_4 \times L_5$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1-\mu)^2/\sigma^2} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(4-\mu)^2/\sigma^2}$$

$$\times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(5-\mu)^2/\sigma^2} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(6-\mu)^2/\sigma^2}$$

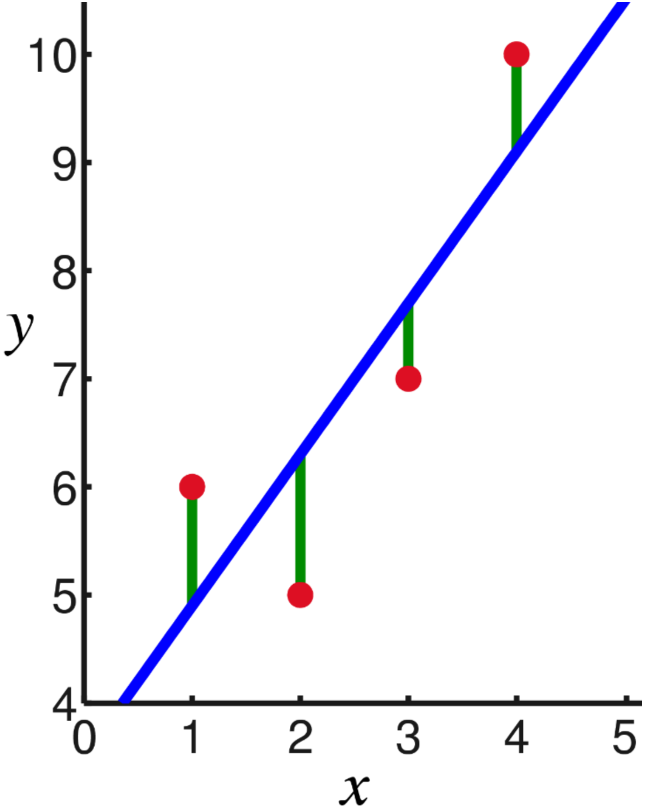
$$\times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(9-\mu)^2/\sigma^2}$$

- The likelihood function depends on mean μ and variance σ^2

The illustration of the estimation procedure

- The value of mean μ and σ that maximise the likelihood function is found.
- The values of mean μ and σ which are obtained this way are called the maximum likelihood estimators of mean μ and σ
- Most of the MLE cannot be solved ‘by hand’. Thus, you need to write an iterative procedure to solve it on computer

Method of Least-squares vs MLE



Model for the expectation (fixed part of the model):

$$E[Y_i] = \beta_0 + \beta_1 x_i$$

Residuals: $r_i = y_i - E[Y_i]$

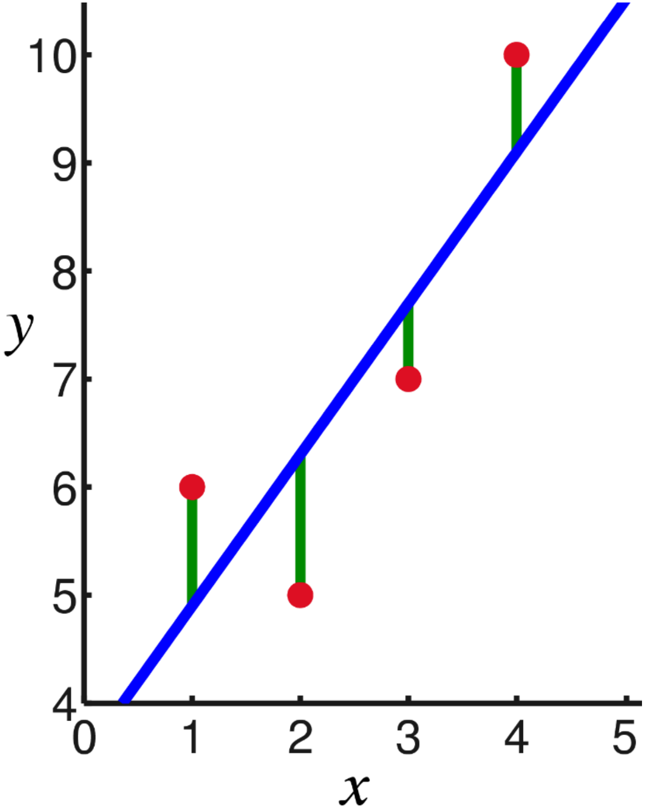
The method of least-squares:

Find the values for the parameters (β_0 and β_1) that makes the sum of the squared residuals (Σr_j^2) as small as possible.

Can only be used when the error term is normal (residuals are assumed to be drawn from a normal distribution)

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \text{ where } \varepsilon_i \sim N(0, \sigma)$$

Method of Least-squares vs MLE



Model for the expectation
(fixed part of the model):

$$E[Y_i] = \beta_0 + \beta_1 x_i$$

Residuals: $r_i = y_i - E[Y_i]$

The maximum likelihood method is more general!

- Can be applied to models with any probability distribution

Estimation of Regression Parameter

- We are interested in estimating a model like this:

$$y = \beta_0 + \beta_1 x + u$$

- Estimating such a model can be done using MLE

Estimation of Regression Parameters

- Suppose that we have the following data and we are interested in estimating the model: $y=\beta_0+\beta_1 x+u$
- Let us make an assumption that u follows the normal distribution with mean 0 and variance σ^2

Id	Y	X
1	2	1
2	6	4
3	7	5
4	9	6
5	15	9

Estimation of Regression Parameters

- We can write the model as :

  u = $y-(\beta_0+\beta_1 x)$

- This means that $y-(\beta_0+\beta_1 x)$ follows the normal distribution with mean 0 and variance σ^2
- The likelihood contribution of each data point is the height of the density function at the data points $(y-\beta_0-\beta_1 x)$

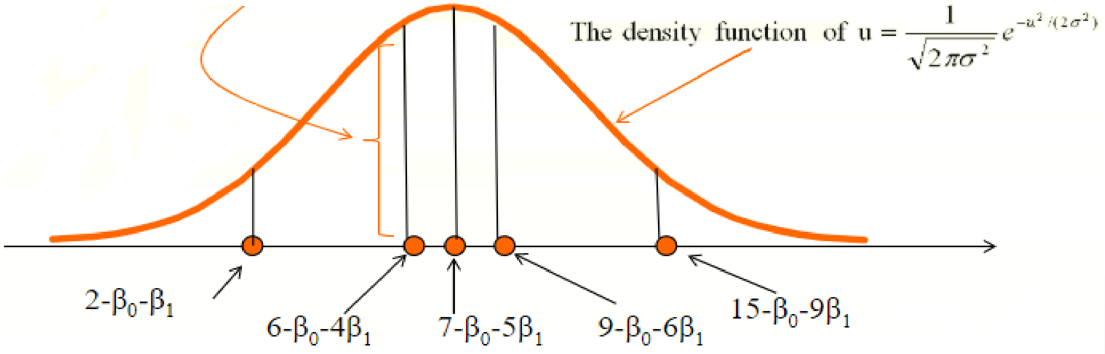
Estimation of Regression Parameters

- The likelihood contribution in this example, of the 2nd observation is given by:

$$L_2 = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(6-\beta_0-4\beta_1)^2/2\sigma^2}$$

Data point

The likelihood contribution of the 2nd observation



Estimation of Regression Parameters

- Then the likelihood function is given by

Id	Y	X
1	2	1
2	6	4
3	7	5
4	9	6
5	15	9

$$\mathrm{L}(\beta_0, \beta_1, \sigma) = \prod_{i=1}^{n} L_i = L_1 \times L_2 \times L_3 \times L_4 \times L_5$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\,(2-\beta_0-\beta_1)^2 / \sigma^2} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\,(6-\beta_0-4\beta_1)^2 / \sigma^2}$$

$$\times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\,(7-\beta_0-5\beta_1)^2 / \sigma^2} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\,(9-\beta_0-6\beta_1)^2 / \sigma^2}$$

$$\times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\,(15-\beta_0-9\beta_1)^2 / \sigma^2}$$

- The likelihood function is a function of β_0 , β_1 and σ.

Estimation of Regression Parameters

- You choose the values of β_0 , β_1 and σ that maximizes the likelihood function.

Python Demo for MLE

In [1]:

```
import numpy as np
from scipy.optimize import minimize
import scipy.stats as stats
```

In [2]:

```
import pandas as pd
tbl = pd.read_excel('mle.xlsx')
tbl
```

Out[2]:

	Id	Y	X
0	1	2	1
1	2	6	4
2	3	7	5
3	4	9	6
4	5	15	9

In [3]:

```python
import statsmodels.api as sm
x= tbl['X']
y =tbl['Y']
x2 = sm.add_constant(x)
modl = sm.OLS(y,x2)
modl2 = modl.fit()
print(modl2.summary())
```

```
C:\Users\HP\Anaconda3\lib\site-packages\statsmodels\compat\pandas.py:56: FutureWarning: The
recated and will be removed in a future version. Please use the pandas.tseries module inste
  from pandas.core import datetools
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      Y   R-squared:                       0.980
Model:                            OLS   Adj. R-squared:                  0.973
Method:                 Least Squares   F-statistic:                     145.9
Date:                Wed, 11 Sep 2019   Prob (F-statistic):            0.00122
Time:                        10:05:16   Log-Likelihood:                -4.5811
No. Observations:                   5   AIC:                             13.16
Df Residuals:                       3   BIC:                             12.38
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.2882      0.755     -0.382      0.728      -2.692       2.115
X              1.6176      0.134     12.079      0.001       1.191       2.044
==============================================================================
Omnibus:                          nan   Durbin-Watson:                   1.405
Prob(Omnibus):                    nan   Jarque-Bera (JB):                0.551
Skew:                           0.089   Prob(JB):                        0.759
Kurtosis:                       1.384   Cond. No.                         12.5
==============================================================================
```

$$b0 = -0.2882 \text{ and } b1 = 1.6176$$

Parameter estimation by MLE

```
In [9]: e=modl2.resid

In [10]: e

Out[10]: 0     0.670588
         1    -0.182353
         2    -0.800000
         3    -0.417647
         4     0.729412
         dtype: float64

In [6]: np.std(e)

Out[6]: 0.6048820983804831
```

Parameter estimation by MLE

In [11]:
```
import numpy as np
from scipy.optimize import minimize
import matplotlib.pyplot as plt

def lik(parameters):
    m = parameters[0]
    b = parameters[1]
    sigma = parameters[2]
    for i in np.arange(0, len(x)):
        y_exp = m * x + b
    L = (len(x)/2 * np.log(2 * np.pi) + len(x)/2 * np.log(sigma ** 2) + 1 /
      (2 * sigma ** 2) * sum((y - y_exp) ** 2))
    return L

x = np.array([1,4,5,6,9])
y = np.array([2,6,7,9,15])
lik_model = minimize(lik, np.array([2,2,2]), method='L-BFGS-B')
```

In [12]:
```
lik_model
```

Out[12]:
```
      fun: 4.581084072762135
 hess_inv: <3x3 LbfgsInvHessProduct with dtype=float64>
      jac: array([1.24344979e-06, 2.84217094e-06, 1.33226763e-06])
  message: b'CONVERGENCE: NORM_OF_PROJECTED_GRADIENT_<=_PGTOL'
     nfev: 108
      nit: 17
   status: 0
  success: True
        x: array([ 1.61764689, -0.28823426,  0.60488214])
```

Example 2

In [1]:
```
import numpy as np
from scipy.optimize import minimize
import scipy.stats as stats
```

In [2]:
```
import pandas as pd
tbl = pd.read_excel('regcar.xlsx')
tbl
```

Out[2]:

	TV Ads	car Sold
0	1	14
1	3	24
2	2	18
3	1	17
4	3	27

```
In [3]: import statsmodels.api as sm
        x= tbl['TV Ads']
        y =tbl['car Sold']
        x2 = sm.add_constant(x)
        modl = sm.OLS(y,x2)
        modl2 = modl.fit()
        print(modl2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               car Sold   R-squared:                       0.877
Model:                            OLS   Adj. R-squared:                  0.836
Method:                 Least Squares   F-statistic:                     21.43
Date:                Wed, 11 Sep 2019   Prob (F-statistic):             0.0190
Time:                        11:11:23   Log-Likelihood:                -9.6687
No. Observations:                   5   AIC:                             23.34
Df Residuals:                       3   BIC:                             22.56
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         10.0000      2.366      4.226      0.024       2.469      17.531
TV Ads         5.0000      1.080      4.629      0.019       1.563       8.437
==============================================================================
Omnibus:                          nan   Durbin-Watson:                   1.214
Prob(Omnibus):                    nan   Jarque-Bera (JB):                0.674
Skew:                           0.256   Prob(JB):                        0.714
Kurtosis:                       1.276   Cond. No.                         6.33
==============================================================================
```

b0 = 10 and b1 = 5

```
In [4]: e=modl2.resid
```

```
In [5]: e
```

```
Out[5]: 0   -1.0
        1   -1.0
        2   -2.0
        3    2.0
        4    2.0
        dtype: float64
```

```
In [9]: np.std(e)
```

```
Out[9]: 1.6733200530681507
```

```
In [10]: import numpy as np
         from scipy.optimize import minimize
         import matplotlib.pyplot as plt

         def lik(parameters):
             m = parameters[0]
             b = parameters[1]
             sigma = parameters[2]
             for i in np.arange(0, len(x)):
                 y_exp = m * x + b
             L = (len(x)/2 * np.log(2 * np.pi) + len(x)/2 * np.log(sigma ** 2) + 1 /
                  (2 * sigma ** 2) * sum((y - y_exp) ** 2))
             return L

         x = np.array([1,3,2,1,3])
         y = np.array([14,24,18,17,27])
         lik_model = minimize(lik, np.array([2,2,2]), method='L-BFGS-B')
```

```
In [11]: lik_model
```

```
Out[11]:       fun: 9.668741208976929
          hess_inv: <3x3 LbfgsInvHessProduct with dtype=float64>
               jac: array([-5.32907052e-07, -1.77635684e-07, -2.30926389e-06])
           message: b'CONVERGENCE: NORM_OF_PROJECTED_GRADIENT_<=_PGTOL'
              nfev: 104
               nit: 22
            status: 0
           success: True
                 x: array([ 5.        ,  9.99999989, -1.67332066])
```

Thank you

IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

LOGISTIC REGRESSION - I

Dr A. RAMESH

DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- Building Logistic regression Model
- Python Demo on Logistic Regression

Application

$y = a + b_{p0} x_1 + b_p x_2$

$(0, 1)$

- In many regression applications the dependent variable may only assume two discrete values.
- For instance, a bank might like to develop an estimated regression equation for predicting whether a person will be approved for a credit card or not
- The dependent variable can be coded as *y* =1 if the bank approves the request for a credit card and *y* = 0 if the bank rejects the request for a credit card.
- Using logistic regression we can estimate the probability that the bank will approve the request for a credit card given a particular set of values for the chosen independent variables.

Example

- Let us consider an application of logistic regression involving a direct mail promotion being used by Simmons Stores.
- Simmons owns and operates a national chain of women's apparel stores.
- Five thousand copies of an expensive four-color sales catalog have been printed, and each catalog includes a coupon that provides a $50 discount on purchases of $200 or more.
- The catalogs are expensive and Simmons would like to send them to only those customers who have the highest probability of using the coupon.

Sources: Statistics for Business and Economics,11th Edition by David R. Anderson (Author), Dennis J. Sweeney (Author), Thomas A. Williams (Author)

Variables

- Management thinks that annual spending at Simmons Stores and whether a customer has a Simmons credit card are two variables that might be helpful in predicting whether a customer who receives the catalog will use the coupon.
- Simmons conducted a pilot study using a random sample of 50 Simmons credit card customers and 50 other customers who do not have a Simmons credit card.
- Simmons sent the catalog to each of the 100 customers selected.
- At the end of a test period, Simmons noted whether the customer used the coupon or not?

Data (10 customer out of 100)

	x_1	x_2	y
Customer	Spending	Card	Coupon
1	2.291	1	0
2	3.215	1	0
3	2.135	1	0
4	3.924	0	0
5	2.528	1	0
6	2.473	0	1
7	2.384	0	0
8	7.076	0	0
9	1.182	1	1
10	3.345	0	0

Explanation of Variables

- The amount each customer spent last year at Simmons is shown in thousands of dollars and the credit card information has been coded as 1 if the customer has a Simmons credit card and 0 if not.
- In the Coupon column, a 1 is recorded if the sampled customer used the coupon and 0 if not.

Logistic Regression Equation

- If the two values of the dependent variable *y* are coded as 0 or 1, the value of *E*(*y*) in equation given below provides the *probability* that *y* = 1 given a particular set of values for the independent variables $x_1, x_2, \ldots, x_p$.

LOGISTIC REGRESSION EQUATION

$$E(y) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p}}$$

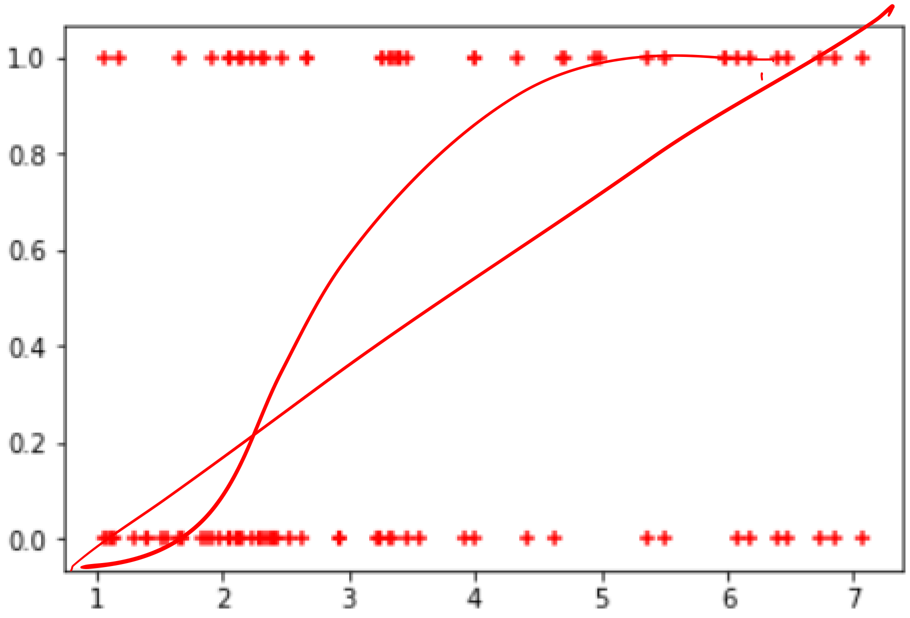
Logistic Regression Equation

- Because of the interpretation of $E(y)$ as a probability, the **logistic regression equation** is often written as follows

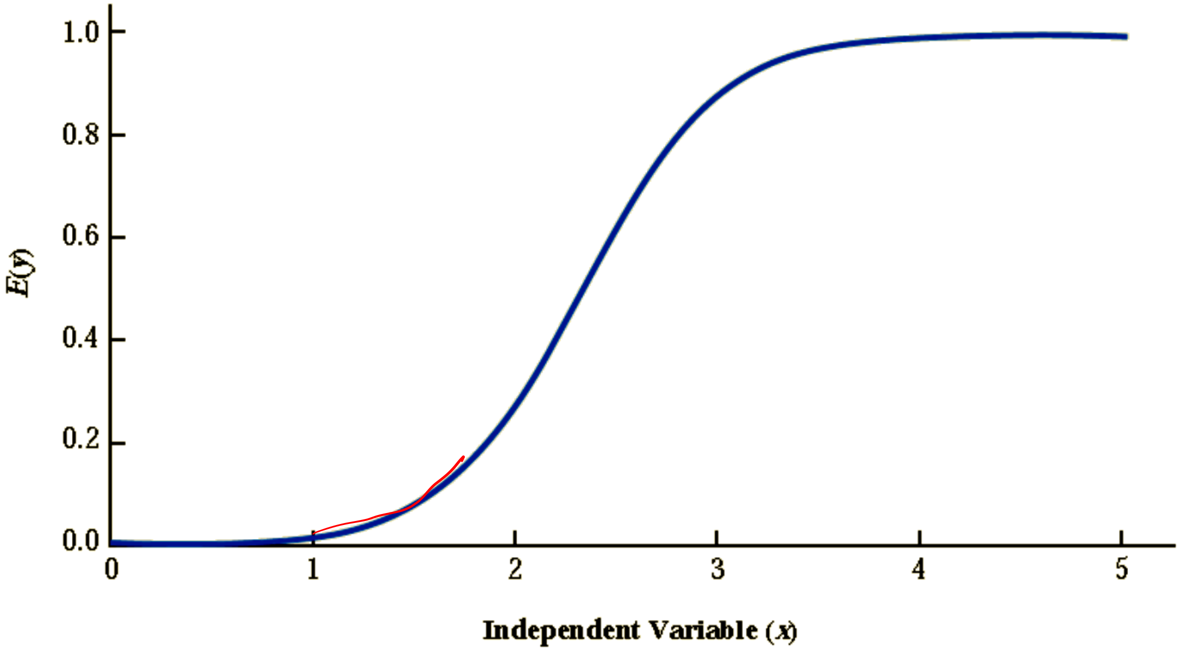
$$E(y) = P(y = 1 | x_1, x_2, \ldots, x_p)$$

In [15]:
```python
plt.scatter(df.Spending,df.Coupon,marker='+',color= 'red')
```

Out[15]: <matplotlib.collections.PathCollection at 0x2a1b5b73c50>



Logistic regression equation for β_0 and β_1



Logistic regression equation for β_0 and β_1

- Note that the graph is S-shaped.
- The value of $E(y)$ ranges from 0 to 1, with the value of $E(y)$ gradually approaching 1 as the value of x becomes larger and the value of $E(y)$ approaching 0 as the value of x becomes smaller.
- Note also that the values of $E(y)$, representing probability, increase fairly rapidly as x increases from 2 to 3.
- The fact that the values of $E(y)$ range from 0 to 1 and that the curve is S-shaped makes equation (slide no.11) ideally suited to model the probability the dependent variable is equal to 1.

Estimating the Logistic Regression Equation

- In simple linear and multiple regression the least squares method is used to compute b_0, b_1, . . . , b_p as estimates of the model parameters (β_0, β_1, . . . , β_p).
- The nonlinear form of the logistic regression equation makes the method of computing estimates more complex MLE
- We will use computer software to provide the estimates.
- The **estimated logistic regression equation** is

ESTIMATED LOGISTIC REGRESSION EQUATION

$$\hat{y} = \text{estimate of } P(y = 1 | x_1, x_2, \ldots, x_p) = \frac{e^{b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_p x_p}}{1 + e^{b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_p x_p}}$$

Here, y hat provides an estimate of the probability that $y = 1$, given a particular set of values for the independent variables.

Python Code for Logistic Regression

In [12]:
```
x = df[['Card','Spending']]
y = df['Coupon']

import statsmodels.api as sm
x1= sm.add_constant(x)
logit_model=sm.Logit(y,x1)
result=logit_model.fit()
print(result.summary2())
```

```
Optimization terminated successfully.
         Current function value: 0.604869
         Iterations 5
                        Results: Logit
=================================================================
Model:              Logit            No. Iterations:   5.0000
Dependent Variable: Coupon           Pseudo R-squared: 0.101
Date:               2019-09-11 12:54 AIC:              126.9739
No. Observations:   100              BIC:              134.7894
Df Model:           2                Log-Likelihood:   -60.487
Df Residuals:       97               LL-Null:          -67.301
Converged:          1.0000           Scale:            1.0000
------------------------------------------------------------------
             Coef.   Std.Err.    z     P>|z|    [0.025   0.975]
------------------------------------------------------------------
const       -2.1464    0.5772 -3.7183 0.0002  -3.2778  -1.0150
Card         1.0987    0.4447  2.4707 0.0135   0.2271   1.9703
Spending     0.3416    0.1287  2.6551 0.0079   0.0894   0.5938
=================================================================
```

Wald

$x_2 \rightarrow$ const

x_1 Spending

G

Variables

$$y = \begin{cases} 0 \text{ if the customer did not use the coupon} \\ 1 \text{ if the customer used the coupon} \end{cases}$$

$$x_1 = \text{annual spending at Simmons Stores (\$1000s)}$$

$$x_2 = \begin{cases} 0 \text{ if the customer does not have a Simmons credit card} \\ 1 \text{ if the customer has a Simmons credit card} \end{cases}$$

$$E(y) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2}}$$

$$\hat{y} = \frac{e^{b_0 + b_1 x_1 + b_2 x_2}}{1 + e^{b_0 + b_1 x_1 + b_2 x_2}} = \frac{e^{-2.14637 + 0.341643 x_1 + 1.09873 x_2}}{1 + e^{-2.14637 + 0.341643 x_1 + 1.09873 x_2}}$$

Managerial Use

- $P(y = 1/x_1 = 2, x_2 = 0) = .1880$

$$\hat{y} = \frac{e^{-2.14637+0.341643(2)+1.09873(0)}}{1 + e^{-2.14637+0.341643(2)+1.09873(0)}} = \frac{e^{-1.4631}}{1 + e^{-1.4631}} = \frac{.2315}{1.2315} = 0.1880$$

- $P(y = 1/x_1= 2,/ x_2 = 1) = .4099$

$$\hat{y} = \frac{e^{-2.14637+0.341643(2)+1.09873(1)}}{1 + e^{-2.14637+0.341643(2)+1.09873(1)}} = \frac{e^{-0.3644}}{1 + e^{-0.3644}} = \frac{.6946}{1.6946} = 0.4099$$

- Probabilities indicate that for customers with annual spending of \$2000 the presence of a Simmons credit card increases the probability of using the coupon

Managerial Use

- It appears that the probability of using the coupon is much higher for customers with a Simmons credit card.

		Annual Spending						
		$1000	$2000	$3000	$4000	$5000	$6000	$7000
Credit Card	Yes	0.3305	0.4099	0.4943	0.5791	0.6594	0.7315	0.7931
	No	0.1413	0.1880	0.2457	0.3144	0.3922	0.4759	0.5610

Testing for Significance

t, F

$$H_0: \beta_1 = \beta_2 = 0$$

H_a: One or both of the parameters is not equal to zero

G Statistics

- The test for overall significance is based upon the value of a *G* test statistic.
- If the null hypothesis is true, the sampling distribution of *G* follows a chi-square distribution with degrees of freedom equal to the number of independent variables in the model.

F

```
In [12]: x = df[['Card','Spending']]
         y = df['Coupon']

         import statsmodels.api as sm
         x1= sm.add_constant(x)
         logit_model=sm.Logit(y,x1)
         result=logit_model.fit()
         print(result.summary2())
```

```
Optimization terminated successfully.
         Current function value: 0.604869
         Iterations 5
                        Results: Logit
================================================================
Model:              Logit            No. Iterations:   5.0000
Dependent Variable: Coupon           Pseudo R-squared: 0.101
Date:               2019-09-11 12:54 AIC:              126.9739
No. Observations:   100              BIC:              134.7894
Df Model:           2                Log-Likelihood:   -60.487
Df Residuals:       97               LL-Null:          -67.301
Converged:          1.0000           Scale:            1.0000
-----------------------------------------------------------------
             Coef.   Std.Err.    z     P>|z|    [0.025   0.975]
-----------------------------------------------------------------
const       -2.1464   0.5772  -3.7183  0.0002  -3.2778  -1.0150
Card         1.0987   0.4447   2.4707  0.0135   0.2271   1.9703
Spending     0.3416   0.1287   2.6551  0.0079   0.0894   0.5938
================================================================
```

G Statistics

$$G = -2\ln\left[\frac{(\text{likelihood without the variable})}{(\text{likelihood with the variable})}\right].$$

$$G = 2(-60.487 - (-67.301) = \boxed{13.628}$$

- The value of G is 13.628, its degrees of freedom are 2,and its p-value is 0.001.
- Thus, at any level of significance $\alpha >= .001$, we would reject the null hypothesis and conclude that the overall model is significant.

Thank You

LOGISTIC REGRESSION - II

Dr A. RAMESH

DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- Testing the significance of Logistic regression coefficients
- Python Demo on Logistic Regression

Chi sq. value of G- Statistic

In [5]:
```
import scipy
from scipy.stats import chi2
```

In [7]:
```
chi2.pdf(13.628,2)
```

Out[7]: 0.000549145469075383

z test- Wald Test

- *z* test can be used to determine whether each of the individual independent variables is making significant contribution to the overall model

$$W = \frac{\hat{\beta}_1}{\widehat{\mathrm{SE}}(\hat{\beta}_1)}:$$

In [12]:
```python
x = df[['Card','Spending']]
y = df['Coupon']

import statsmodels.api as sm
x1= sm.add_constant(x)
logit_model=sm.Logit(y,x1)
result=logit_model.fit()
print(result.summary2())
```

```
Optimization terminated successfully.
         Current function value: 0.604869
         Iterations 5
                        Results: Logit
=================================================================
Model:              Logit            No. Iterations:   5.0000
Dependent Variable: Coupon           Pseudo R-squared: 0.101
Date:               2019-09-11 12:54 AIC:              126.9739
No. Observations:   100              BIC:              134.7894
Df Model:           2                Log-Likelihood:   -60.487
Df Residuals:       97               LL-Null:          -67.301
Converged:          1.0000           Scale:            1.0000
------------------------------------------------------------------
              Coef.   Std.Err.     z     P>|z|    [0.025   0.975]
------------------------------------------------------------------
const        -2.1464    0.5772  -3.7183  0.0002  -3.2778  -1.0150
Card          1.0987    0.4447   2.4707  0.0135   0.2271   1.9703
Spending      0.3416    0.1287   2.6551  0.0079   0.0894   0.5938
=================================================================
```

Strategies

- Suppose Simmons wants to send the promotional catalog only to customers who have a 0.40 or higher probability of using the coupon.
- **Customers who have a Simmons credit card:** Send the catalog to every customer who spent $2000 or more last year.
- **Customers who do not have a Simmons credit card:** Send the catalog to every customer who spent $6000 or more last year.

		Annual Spending						
		$1000	$2000	$3000	$4000	$5000	$6000	$7000
Credit Card	Yes	0.3305	0.4099	0.4943	0.5791	0.6594	0.7315	0.7931
	No	0.1413	0.1880	0.2457	0.3144	0.3922	0.4759	0.5610

Interpreting the Logistic Regression Equation

$$\text{odds} = \frac{P(y = 1|x_1, x_2, \ldots, x_p)}{P(y = 0|x_1, x_2, \ldots, x_p)} = \frac{P(y = 1|x_1, x_2, \ldots, x_p)}{1 - P(y = 1|x_1, x_2, \ldots, x_p)}$$

Odd ratio

$$\text{Odds Ratio} = \frac{\text{odds}_1}{\text{odds}_0}$$

- The **odds ratio** measures the impact on the odds of a one-unit increase in only one of the independent variables.

Interpretation

- For example, suppose we want to compare the odds of using the coupon for customers who spend $2000 annually and have a Simmons credit card ($x1 = 2$ and $x2 = 1$) to the odds of using the coupon for customers who spend $2000 annually and do not have a Simmons credit card ($x1 = 2$ and $x2 = 0$).
- We are interested in interpreting the effect of a one-unit increase in the independent variable $x2$.

Odds ratio

$$\text{odds}_1 = \frac{P(y = 1|x_1 = 2, x_2 = 1)}{1 - P(y = 1|x_1 = 2, x_2 = 1)}$$

$$\text{estimate of odds}_1 = \frac{.4099}{1 - .4099} = .6946$$

$$\text{odds}_0 = \frac{P(y = 1|x_1 = 2, x_2 = 0)}{1 - P(y = 1|x_1 = 2, x_2 = 0)}$$

$$\text{estimate of odds}_0 = \frac{.1880}{1 - .1880} = .2315$$

$$\text{Estimated odds ratio} = \frac{.6946}{.2315} = 3.00$$

		Annual Spending						
		\$1000	\$2000	\$3000	\$4000	\$5000	\$6000	\$7000
Credit Card	Yes	0.3305	0.4099	0.4943	0.5791	0.6594	0.7315	0.7931
	No	0.1413	0.1880	0.2457	0.3144	0.3922	0.4759	0.5610

Odds ratio – Interpretation

- The estimated odds in favor of using the coupon for customers who spent \$2000 last year and have a Simmons credit card are 3 times greater than the estimated odds in favor of using the coupon for customers who spent \$2000 last year and do not have a Simmons credit card.

Odds ratio – Interpretation

- The odds ratio for each independent variable is computed while holding all the other independent variables constant.
- But it does not matter what constant values are used for the other independent variables.
- For instance, if we computed the odds ratio for the Simmons credit card variable ($x2$) using $3000, instead of $2000, as the value for the annual spending variable ($x1$), we would still obtain the same value for the estimated odds ratio (3.00).
- Thus, we can conclude that the estimated odds of using the coupon for customers who have a Simmons credit card are 3 times greater than the estimated odds of using the coupon for customers who do not have a Simmons credit card.

Relationship between the odds ratio and the coefficients of the independent variables

$$\text{Odds ratio} = e^{\beta_i}$$

		Annual Spending						
		\$1000	\$2000	\$3000	\$4000	\$5000	\$6000	\$7000
Credit Card	Yes	0.3305	0.4099	0.4943	0.5791	0.6594	0.7315	0.7931
	No	0.1413	0.1880	0.2457	0.3144	0.3922	0.4759	0.5610

$$\text{Estimated odds ratio} = e^{b_1} = e^{.341643} = 1.41$$

ted odds ratio for x_2 is

$$\text{Estimated odds ratio} = e^{b_2} = e^{1.09873} = 3.00$$

Effect of a change of more than one unit in Odd Ratio

$x_1 = 2$ $x_1 = 3$ $x_2 = 0$ $x_2 = 1$

- The odds ratio for an independent variable represents the change in the odds for a one unit change in the independent variable holding all the other independent variables constant.
- Suppose that we want to consider the effect of a change of more than one unit, say c units.
- For instance, suppose in the Simmons example that we want to compare the odds of using the coupon for customers who spend \$5000 annually ($x_1 = 5$) to the odds of using the coupon for customers who spend \$2000 annually ($x_1 = 2$).
- In this case $c = 5 - 2 = 3$ and the corresponding estimated odds ratio is

Effect of a change of more than one unit in Odd Ratio

$$e^{cb_1} = e^{3(.341643)} = e^{1.0249} = 2.79$$

- This result indicates that the estimated odds of using the coupon for customers who spend $5000 annually is 2.79 times greater than the estimated odds of using the coupon for customers who spend $2000 annually.
- In other words, the estimated odds ratio for an increase of $3000 in annual spending is 2.79

Logit Transformation

- An interesting relationship can be observed between the odds in favor of $y = 1$ and the exponent for 'e' in the logistic regression equation

$$\ln(\text{odds}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

- This equation shows that the natural logarithm of the odds in favor of $y = 1$ is a linear function of the independent variables.
- This linear function is called the **logit** →$g(x1, x2, . . . , xp)$ to denote the logit.

Estimated Logit Regression Equation

$$g(x_1, x_2, \ldots, x_p) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

$$E(y) = \frac{e^{g(x_1, x_2, \ldots, x_p)}}{1 + e^{g(x_1, x_2, \ldots, x_p)}}$$

$$\hat{y} = \frac{e^{b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_p x_p}}{1 + e^{b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_p x_p}} = \frac{e^{\hat{g}(x_1, x_2, \ldots, x_p)}}{1 + e^{\hat{g}(x_1, x_2, \ldots, x_p)}}$$

$$\hat{g}(x_1, x_2) = -2.14637 + 0.341643x_1 + 1.09873x_2$$

$$\hat{y} = \frac{e^{\hat{g}(x_1, x_2)}}{1 + e^{\hat{g}(x_1, x_2)}} = \frac{e^{-2.14637 + 0.341643x_1 + 1.09873x_2}}{1 + e^{-2.14637 + 0.341643x_1 + 1.09873x_2}}$$

G vs Z

- Because of the unique relationship between the estimated coefficients in the model and the corresponding odds ratios, the overall test for significance based upon the G statistic is also a test of overall significance for the odds ratios.
- In addition, the z test for the individual significance of a model parameter also provides a statistical test of significance for the corresponding odds ratio.

Thank You

IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Linear Regression Model Vs Logistic Regression Model

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT STUDIES

Agenda

- Comparison of Linear Regression model and Logistic regression model

Estimating the relationship

Linear regression model

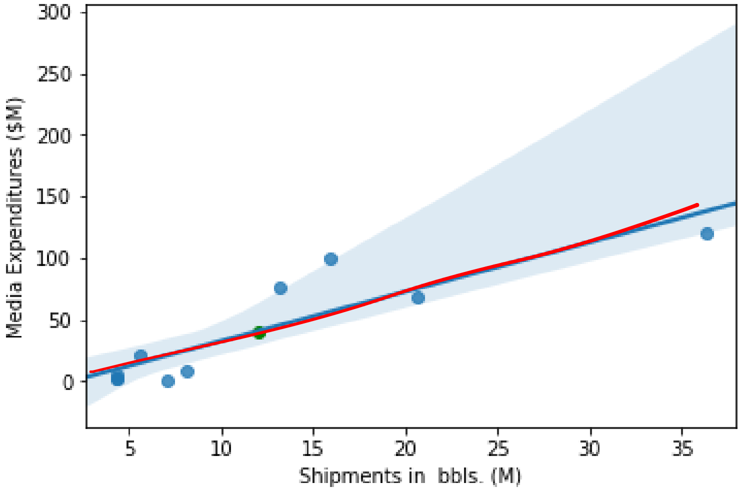
- $Y_1 = X_1 + X_2 + \ldots + X_n$
- Where ,
  - Y_1 = continuous data
  - Independent variables = nonmetric and metric

Logistic regression model

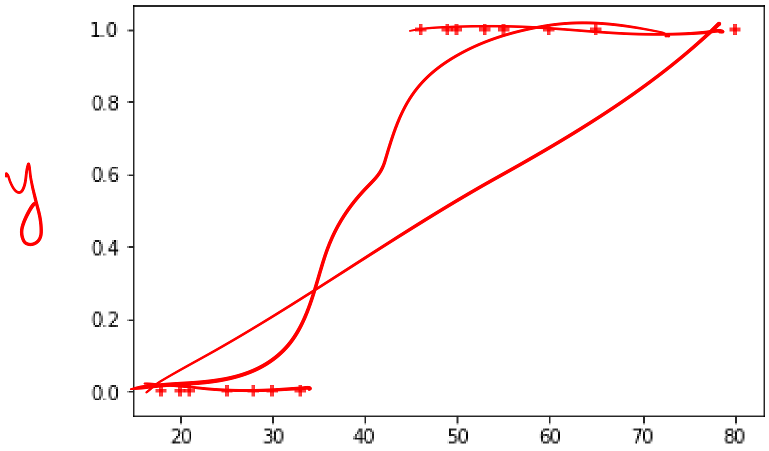
- $Y_1 = X_1 + X_2 + \ldots + X_n$
- Where ,
  - Y_1 = Binary nonmetric
  - Independent variables = nonmetric and metric

Graphical representation

- **Linear regression**



- **Logistic regression**



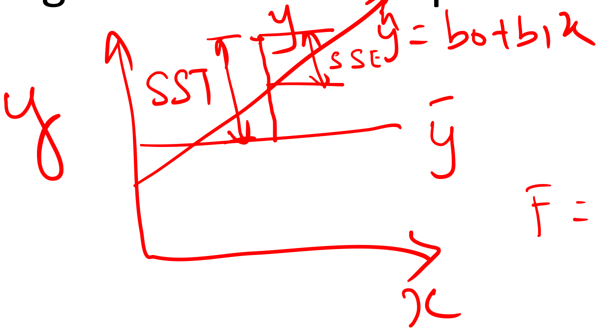
Correspondence of Primary Elements of Model Fit

Linear Regression

- Total sum of squares SST
- Error sum of squares SSE
- F test of model fit
- Coefficient of determination (R^2)
- Regression sum of squares SSR

Logistic Regression

- -2LL of base model
- -2LL of proposed model
- Chi-square test of -2LL difference (G)
- Pseudo R^2 measures
- Difference of -2LL for base and proposed models



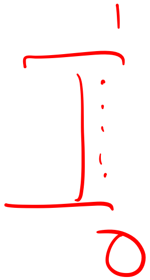
$$F = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n-k-1)} \qquad R^2 = \frac{SSR}{SST}$$

Objective of logistic regression

- Logistic regression is identical to discriminant analysis in terms of the basic objectives it can address
- Logistic regression is best suited to address two research objectives:
  - Identifying the independent variables that impact group membership in the dependent variable
  - Establishing a classification system based on the logistic model for determining group membership

The fundamental difference

- Logistic regression differs from linear regression, in being specifically designed to predict the probability of an event occurring (ie., the probability of an observation being in the group coded 1)
- Although probability values are metric measures, there are fundamental differences between linear regression and logistic regression



Log likelihood

- Measure used in logistic regression to represent the lack of predictive fit
- Even though this method does not use the least squares procedure in model estimation, as is done in linear regression, the likelihood value is similar to the sum of squared error in regression analysis

SSE

Logistic vs discriminant

- Logistic regression may be preferred for two reasons
- First, discriminant analysis relies on strictly meeting the assumptions of
  - Multivariate normality and equal variance
  - Covariance matrices across groups
  - Assumptions that are not met in many situations
- Logistic regression does not face these strict assumptions and is much more robust when these assumptions are not met, making its application appropriate in many situations

Logistic vs discriminant

- Second, even if the assumptions are met, many researchers prefer logistic regression because it is similar to multiple regression
- It has straightforward statistical tests, similar approaches to incorporating metric and nonmetric variables and nonlinear effects, and a wide range of diagnostics
- Logistic regression is equivalent to two-group discriminant analysis and may be more suitable in many situations

Logistic vs discriminant : Sample size

- One factor that distinguishes logistic regression from the other techniques is its use of maximum likelihood (MLE) as the estimation technique
- MLE requires larger samples such that, all things being equal, logistic regression will require a larger sample size than multiple regression
- As for discriminant analysis, there are considerations on the minimum group size as well

Logistic vs discriminant : Sample size

- The recommended sample size for each group is at least 10 observations per estimated parameter 3 –
- This is much greater than multiple regression, which had a minimum of five observations per parameter, and that was for the overall sample, not the sample size for each group, as seen with logistic regression

Regression : 1 – 5

L. R : 1 : 10

Determination of coefficients

Linear regression

- R^2
  - r^2 = SSR/SST

  where:

  SSR = sum of squares due to regression

  SST = total sum of squares

Logistic regression

$$R^2 Logit = \frac{-2LLnull - (-2LLmodel)}{-2LLnull}$$

Where:

LL = Loglikelihood

$-2LL_{null}$ = -2LL of base model

$-2LL_{model}$= -2LL of proposed model

Determination of coefficients

- **Linear regression**

OLS Regression Results

Dep. Variable:	Media Expenditures ($M)	**R-squared:**	0.783
Model:	OLS	**Adj. R-squared:**	0.756
Method:	Least Squares	**F-statistic:**	28.93
Date:	Wed, 10 Oct 2018	**Prob (F-statistic):**	0.000663
Time:	18:16:26	**Log-Likelihood:**	-44.355
No. Observations:	10	**AIC:**	92.71
Df Residuals:	8	**BIC:**	93.32
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-7.6277	11.485	-0.664	0.525	-34.112	18.857
Shipments in bbls. (M)	4.0065	0.745	5.378	0.001	2.289	5.724

Omnibus:	4.361	**Durbin-Watson:**	1.473
Prob(Omnibus):	0.113	**Jarque-Bera (JB):**	2.129
Skew:	1.129	**Prob(JB):**	0.345
Kurtosis:	2.925	**Cond. No.**	24.6

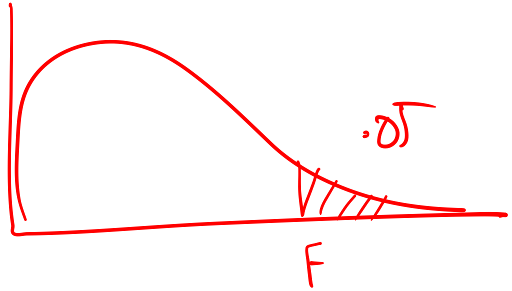
- **Logistic regression**

```
                          Results: Logit
===============================================================
Model:              Logit              Pseudo R-squared: 0.192
Dependent Variable: Coupon             AIC:              12.0864
Date:               2019-09-08 11:07   BIC:              12.6916
No. Observations:   10                 Log-Likelihood:   -4.0432
Df Model:           1                  LL-Null:          -5.0040
Df Residuals:       8                  LLR p-value:      0.16568
Converged:          1.0000             Scale:            1.0000
No. Iterations:     7.0000
---------------------------------------------------------------
             Coef.   Std.Err.     z     P>|z|    [0.025  0.975]
---------------------------------------------------------------
Spending    -0.6318   0.4566  -1.3838  0.1664  -1.5267  0.2630
Card        -0.0029   1.4887  -0.0020  0.9984  -2.9207  2.9149
===============================================================
```

Testing for overall significance

Linear regression

- F-test of model fit
- F = MSR/MSE

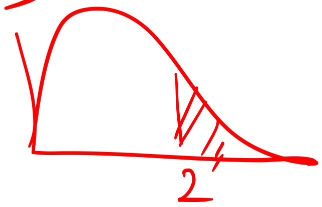


- **Logistic Regression**
- G-test of model fit

$$G = -2\ln\left[\frac{likelihood\ \ without\ \ the\ \ variable}{likelihood\ \ with\ \ variable}\right]$$

$$= -2[-5-(-4)]$$

$$= -2[-1] = 2$$



Testing for overall significance

- **Linear regression**

OLS Regression Results

Dep. Variable:	Media Expenditures ($M)	R-squared:	0.783
Model:	OLS	Adj. R-squared:	0.756
Method:	Least Squares	F-statistic:	28.93
Date:	Wed, 10 Oct 2018	Prob (F-statistic):	0.000663
Time:	18:16:26	Log-Likelihood:	-44.355
No. Observations:	10	AIC:	92.71
Df Residuals:	8	BIC:	93.32
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-7.6277	11.485	-0.664	0.525	-34.112	18.857
Shipments in bbls. (M)	4.0065	0.745	5.378	0.001	2.289	5.724

Omnibus:	4.361	Durbin-Watson:	1.473
Prob(Omnibus):	0.113	Jarque-Bera (JB):	2.129
Skew:	1.129	Prob(JB):	0.345
Kurtosis:	2.925	Cond. No.	24.6

- **Logistic regression**

```
                         Results: Logit
==============================================================
Model:              Logit            Pseudo R-squared: 0.192
Dependent Variable: Coupon           AIC:              12.0864
Date:               2019-09-08 11:07 BIC:              12.6916
No. Observations:   10               Log-Likelihood:   -4.0432
Df Model:           1                LL-Null:          -5.0040
Df Residuals:       8                LLR p-value:      0.16568
Converged:          1.0000           Scale:            1.0000
No. Iterations:     7.0000
---------------------------------------------------------------
            Coef.   Std.Err.     z     P>|z|    [0.025  0.975]
---------------------------------------------------------------
Spending   -0.6318   0.4566  -1.3838  0.1664  -1.5267  0.2630
Card       -0.0029   1.4887  -0.0020  0.9984  -2.9207  2.9149
==============================================================
```

Testing for significance

Linear regression

- t-test

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

$$t = \frac{b_1 - \beta_1}{S_b}$$

$$\text{where:} \quad S_b = \frac{S_e}{\sqrt{SS_{XX}}}$$

$$S_e = \sqrt{\frac{SSE}{n-2}}$$

$$SS_{XX} = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$\beta_1 = \text{the hypothesized slope}$$

$$df = n - 2$$

Logistic regression

- Wald-test

$$W = \frac{\hat{\beta}_1}{\widehat{SE}(\hat{\beta}_1)} = \frac{-0.0029}{1.4882}$$

$=$

Testing for significance

- **Linear regression**

OLS Regression Results

Dep. Variable:	Media Expenditures ($M)	R-squared:	0.783
Model:	OLS	Adj. R-squared:	0.756
Method:	Least Squares	F-statistic:	28.93
Date:	Wed, 10 Oct 2018	Prob (F-statistic):	0.000663
Time:	18:16:26	Log-Likelihood:	-44.355
No. Observations:	10	AIC:	92.71
Df Residuals:	8	BIC:	93.32
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-7.6277	11.485	-0.664	0.525	34.112	18.857
Shipments in bbls. (M)	4.0065	0.745	5.378	0.001	2.289	5.724

Omnibus:	4.361	Durbin-Watson:	1.473
Prob(Omnibus):	0.113	Jarque-Bera (JB):	2.129
Skew:	1.129	Prob(JB):	0.345
Kurtosis:	2.925	Cond. No.	24.6

- **Logistic regression**

Results: Logit

Model:	Logit	Pseudo R-squared:	0.192
Dependent Variable:	Coupon	AIC:	12.0864
Date:	2019-09-08 11:07	BIC:	12.6916
No. Observations:	10	Log-Likelihood:	-4.0432
Df Model:	1	LL-Null:	-5.0040
Df Residuals:	8	LLR p-value:	0.16568
Converged:	1.0000	Scale:	1.0000
No. Iterations:	7.0000		

	Coef.	Std.Err.	z	P>|z|	[0.025	0.975]
Spending	-0.6318	0.4566	-1.3838	0.1664	-1.5267	0.2630
Card	-0.0029	1.4887	-0.0020	0.9984	-2.9207	2.9149

Model Estimation fit

- The basic measure of how well the maximum likelihood estimation procedure fits is the likelihood value, similar to the sums of squares values used in multiple regression SSE
- Logistic regression measures model estimation fit with the value of -2 times the log of the likelihood value, referred to as -2LL or -2 log likelihood
- The minimum value for -2LL is 0, which corresponds to a perfect fit (likelihood = 1 and -2LL is then 0)

Model Estimation fit

- The lower the -2LL value, the better the fit of the model
- The -2LL value can be used to compare equations for the change in fit

Between Model Comparison

- The likelihood value can be compared between equations to assess the difference in predictive fit from one equation to another, with statistical tests for the significance of these differences
- The basic approach follows three steps:

Step 1 : Estimate a null model

- The first step is to calculate a null model, which acts as the baseline for making comparisons of improvement in model fit.
- The most common null model is one without any independent variables, which is similar to calculating the total sum of squares using only the mean in linear regression. $\bar{y}$
- The logic behind this form of null model is that it can act as a baseline against which any model containing independent variables can be compared.

Step 2: Estimate the proposed model

- This model contains the independent variables to be included in the logistic regression model.
- This model fit will improve from the null model and result in a lower -2lL value.
- Any number of proposed models can be estimated

Step 3: **Assess -2LL difference**:

- The final step is to assess the statistical significance of the -2LL value between the two models (null model versus proposed model).
- If the statistical tests support significant differences, then we can state that the set of independent variable(s) in the proposed model is significant in improving model estimation fit.

Between model comparison

Linear regression

- SSE
- $= \sum(y_i - \hat{y}_i)^2$

Logistic Regression

- -2LL of proposed model

Between model comparison

Linear Regression

- SSR = = $\sum(y_i \_ \bar{y}_i)^2$
- SST-SSE

Logistic regression

- Difference between log likelihood
- = $2LL_{null} - (2LL_{model})$

Normality of Residual (Error)

Linear regression

- Normally distributed
- Linear regression assumes that residuals are approximately equal for all predicted dependent variable values

Logistic regression

- Binomially distributed
- Logistic regression does not need residuals to be equal for each level of the predicted dependent variable values

Estimation Methods

- Linear regression is based on least square estimation
- Regression coefficients should be chosen in such a way that it minimizes the sum of the squared distances of each observed response to its fitted value

S

- logistic regression is based on Maximum Likelihood Estimation
- Coefficients should be chosen in such a way that it maximizes the Probability of Y given X (likelihood)
- With MLE, the computer uses different "iterations" in which it tries different solutions until it gets the maximum likelihood estimates

Interpretation

Coefficients of linear regression is interpreted as:

- Keeping all other independent variables constant, how much the dependent variable is expected to increase/decrease with an unit increase in the independent variable

In logistic regression, we interpret odd ratios as:

- The effect of a one unit of change in X in the predicted odds ratio with the other variables in the model held constant

3

THANK YOU